External Network (Hospital
Mobile)
Real MRI/X-ray Machine
Doctor's Smartphone
DICOM
HTTP POST
NVIDIA Jetson Nano 4GB
e Device)
Data Entry
pynetdicom C-STORE
Listener
Flask REST API :5000
STAGE 0: OOD Bouncer
Convolutional Autoencoder
MSE > Threshold?
Yes
UNKNOWN PATHOLOGY
Cannot Diagnose
Log Alert
Local SQLite DB
Fetch History if Patient
Exists
The 3-Stage Threaded
Pipel
P
STAGE 3: Post-Process &
Output
(CPU Thread)
MC Dropout / True CAM
Patient Fusion / Phi-2 LLM
No
Confidence-Gated Routing
(Fe re 8)
Evaluate Confidence
& Uncertainty
High Conf
Low Conf / OOD
Medium Conf
Read
Read
TIER 1: Auto-Approve
>90% Conf, <5% Uncert
TIER 3: Urgent Review
<70% Conf OR >10% Uncert
OR OOD
TIER 2: Verification
Recommended
70-90% Conf OR 5-10%
Uncert
Normal Worklist DICOM SR
Flagged Worklist DICOM SR
Urgent Worklist DICOM SR
GPIO Breadboard
Green LED
Red LED
Piezo Buzzer
The Learning Loop
(Features 3, 6, 7)
Active Learning Queue
Display on UI
Orthanc PACS Server
Pull
STAGE 1: Fetch & Preprocess
(CPU Thread)
OpenCV Decode / Resize
224x224
Normalize / CHW format
Push Tensor
Queue #1
Pull
STAGE 2: TensorRT Inference
(GPU Thread)
Memcpy H2D / FP16
execute_v2
Memcpy D2H
Push Logits & Features
Queue #2
EWC Continual Learning
Loss = task + λ × Σ(Fisher ×
Δθ²)
Auto Re-export & Hot-Swap
Hot-swaps updated model
Sync
Federated Learning Server
Aggregates Global Weights
Sync Logs
GUI Dashboard (Always
Running)
Tab 1: Scan Upload & Results
Tab 2: Patient Risk
Dashboard
Tab 3: Active Learning
Queue
Doctor Submits Correction
Tab 4: Hardware Monitor
tegrastats live feed
Tab 5: Federated Learning